```
import numpy as np
import pandas as pd

df = pd.read_csv("Fifa_world_cup_matches.csv")
df
```

| | team1 | team2 | possession team1 | possession team2 | possession in contest | number of goals team1 | number of goals team2 | date | hour | category | ... | penalties scored team1 | penalti scor tea |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | QATAR | ECUADOR | 42% | 50% | 8% | 0 | 2 | 20 NOV 2022 | 17 : 00 | Group A | ... | 0 | |
| **1** | ENGLAND | IRAN | 72% | 19% | 9% | 6 | 2 | 21 NOV 2022 | 14 : 00 | Group B | ... | 0 | |
| **2** | SENEGAL | NETHERLANDS | 44% | 45% | 11% | 0 | 2 | 21 NOV 2022 | 17 : 00 | Group A | ... | 0 | |
| **3** | UNITED STATES | WALES | 51% | 39% | 10% | 1 | 1 | 21 NOV 2022 | 20 : 00 | Group B | ... | 0 | |
| **4** | ARGENTINA | SAUDI ARABIA | 64% | 24% | 12% | 1 | 2 | 22 NOV 2022 | 11 : 00 | Group C | ... | 1 | |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | |
| **59** | ENGLAND | FRANCE | 54% | 36% | 10% | 1 | 2 | 10 DEC 2022 | 20 : 00 | Quarter-final | ... | 1 | |
| **60** | ARGENTINA | CROATIA | 34% | 54% | 12% | 3 | 0 | 13 DEC 2022 | 20 : 00 | Semi-final | ... | 1 | |
| **61** | FRANCE | MOROCCO | 34% | 55% | 11% | 2 | 0 | 14 DEC 2022 | 20 : 00 | Semi-final | ... | 0 | |
| **62** | CROATIA | MOROCCO | 45% | 45% | 10% | 2 | 1 | 17 DEC 2022 | 16 : 00 | Play-off for third place | ... | 0 | |
| **63** | ARGENTINA | FRANCE | 46% | 40% | 14% | 3 | 3 | 18 DEC 2022 | 16 : 00 | Final | ... | 1 | |

64 rows × 88 columns

```
# 1. Correcting the arange operation
# We convert the 'possession in contest' percentage strings to integers first
possession_ints = df['possession in contest'].str.rstrip('%').astype(int).values
a1 = np.arange(len(possession_ints))
print(f"Created an index array of length: {len(a1)}")

# 2. Basic Statistics with Numpy
total_goals_team1 = df['number of goals team1'].values
print(f"Mean goals (Team 1): {np.mean(total_goals_team1):.2f}")
print(f"Max goals (Team 1): {np.max(total_goals_team1)}")

# 3. Vectorized Operations
# Calculate total goals per match using numpy addition
total_goals_per_match = np.add(df['number of goals team1'].values, df['number of goals team2'].values)
df['total_goals'] = total_goals_per_match

# 4. Conditional Selection with np.where
# Identify matches with more than 3 goals
high_scoring = np.where(total_goals_per_match > 3, "High", "Low")
print(f"Count of high scoring matches: {np.sum(high_scoring == 'High')}")

display(df[['team1', 'team2', 'total_goals']].head())
```

```
a8 = df['defensive pressures applied team1'].values
a9 = np.arange(len(a8))
print(a9)
print(f"Max goals (Team 1): {np.max(a8)}")
a11 = np.zeros(len(a8))
print(a11)
```

```
Created an index array of length: 64
Mean goals (Team 1): 1.58
Max goals (Team 1): 7
Count of high scoring matches: 16
```

| | team1 | team2 | total_goals |
|---|---|---|---|
| **0** | QATAR | ECUADOR | 2 |
| **1** | ENGLAND | IRAN | 8 |
| **2** | SENEGAL | NETHERLANDS | 2 |
| **3** | UNITED STATES | WALES | 2 |
| **4** | ARGENTINA | SAUDI ARABIA | 3 |

```
[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47
 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63]
Max goals (Team 1): 637
[0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.
 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.
 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
```

```
mini= df['defensive pressures applied team2'].min()
mini
```

```
141
```

```
maxw= df['defensive pressures applied team2'].max()
maxw
```

```
585
```

```
tea = df['defensive pressures applied team2'].split()
tea
```

```
np.float64(293.265625)
```